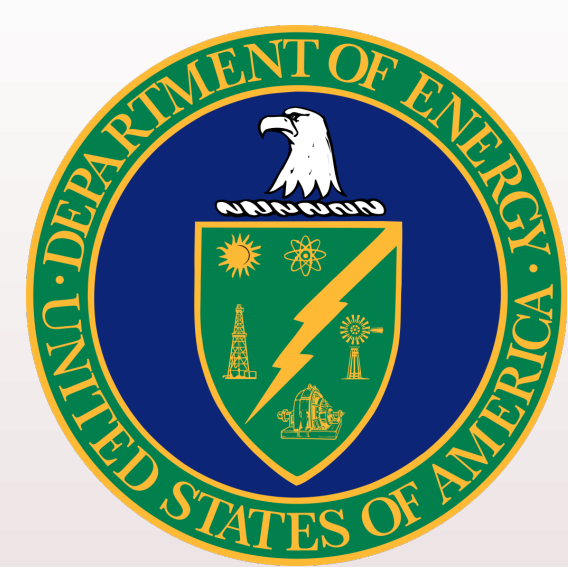




SPECTROSCOPY OF ^{12}Be USING TEXAT TPC

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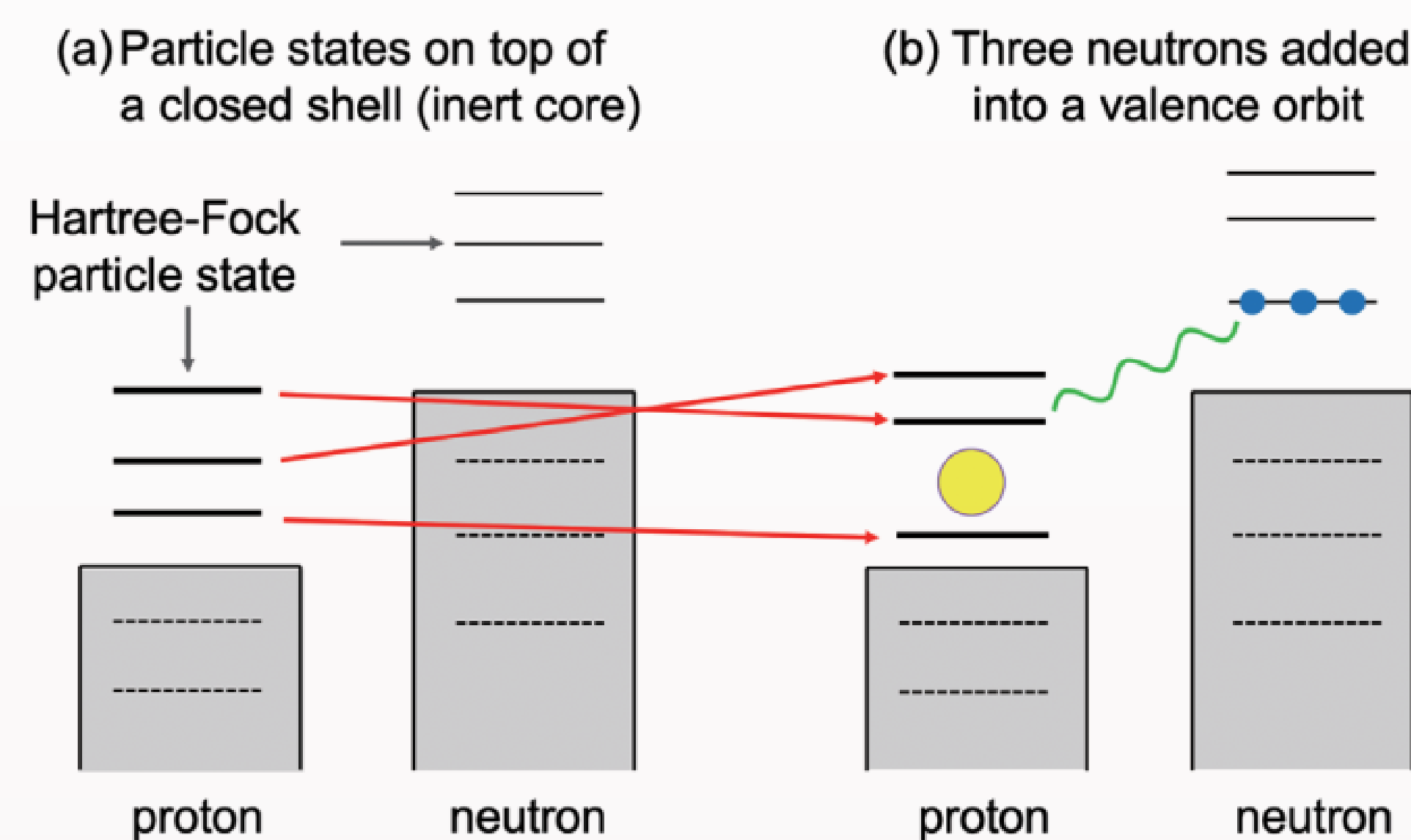


CYCLOTRON INSTITUTE
TEXAS A & M UNIVERSITY

SUMMARY

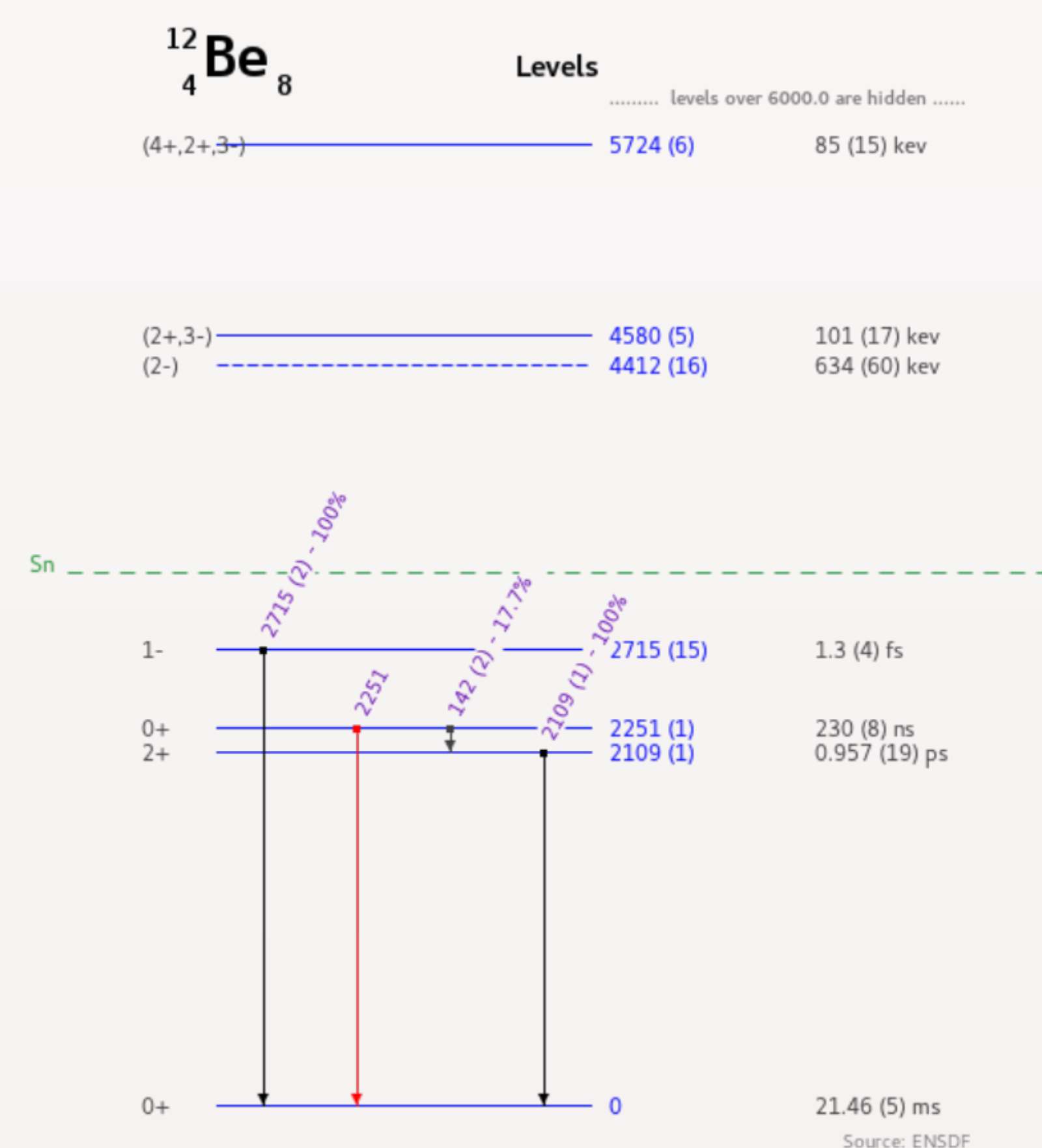
We will use $^{13}\text{B}(d, ^3\text{He})^{12}\text{Be}$ in the TexAT-TPC to determine the proton removal spectroscopic factors (SF's) of low lying states in ^{12}Be .

MOTIVATION



Beryllium isotopes are particularly useful for studying shell evolution as they sit at the intersection of experimental and theoretical (ab-initio) accessibility. [Fig. from 1]

The $\nu(sd)^2$ vs. $\nu(p)^2$ contributions to the low-lying states in ^{12}Be is not settled.[2]



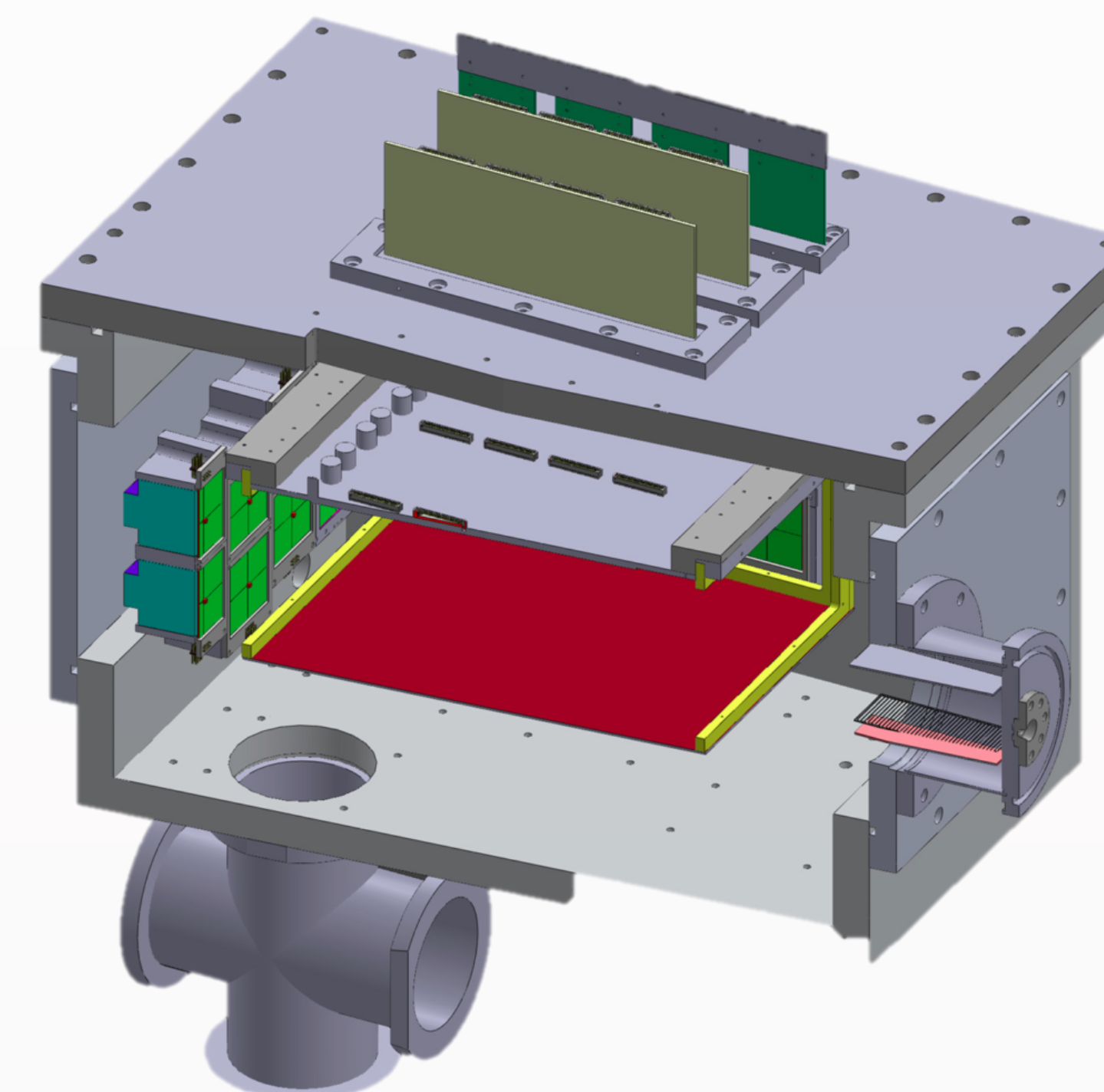
Extracting SF's with $^{13}\text{B}(d, ^3\text{He})^{12}\text{Be}$ will complement existing (d,p) and (t,p) measurements.

TEXAS ACTIVE TARGET (TexAT) DETECTOR [3]

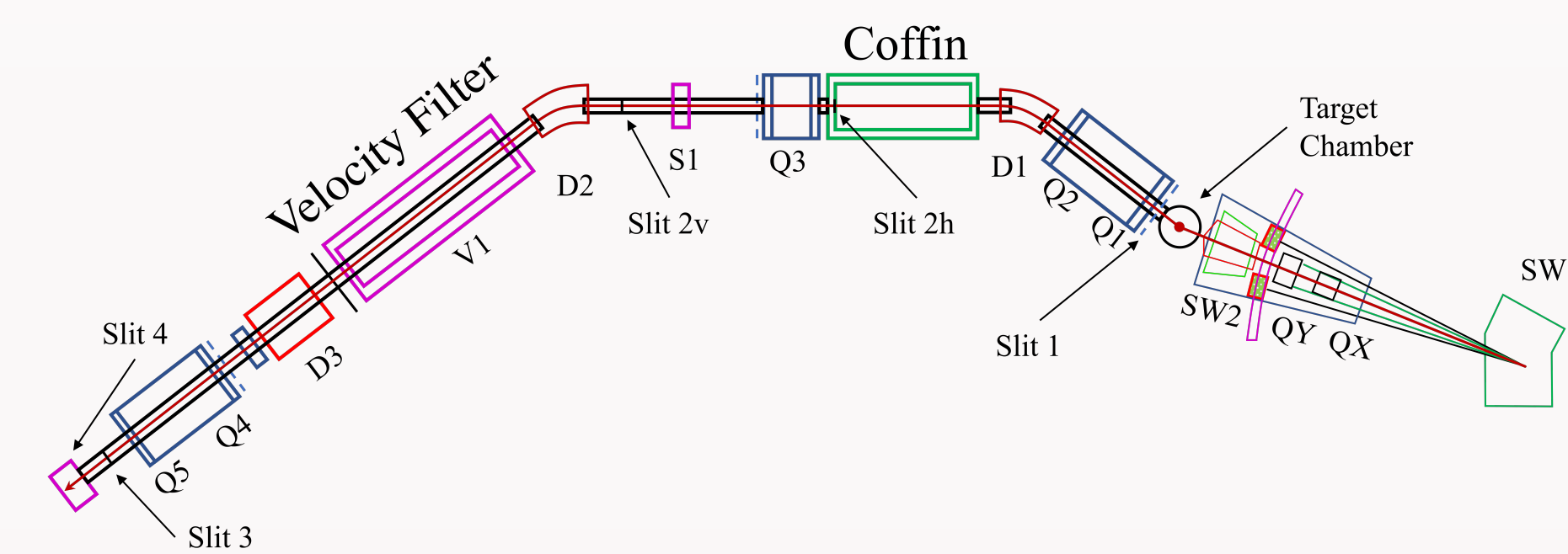
TexAT is an active-target time-projection chamber (TPC), coupled with a solid state detector array. It features:

- TPC based on GEM + Micromegas
- $224 \times 240 \times 135 \text{ mm}^3$ active volume
- $10 \times \text{Si-CsI telescopes}$ (forward wall only)
- 0° Telescope $\rightarrow 16 \times 16$ DSSD.

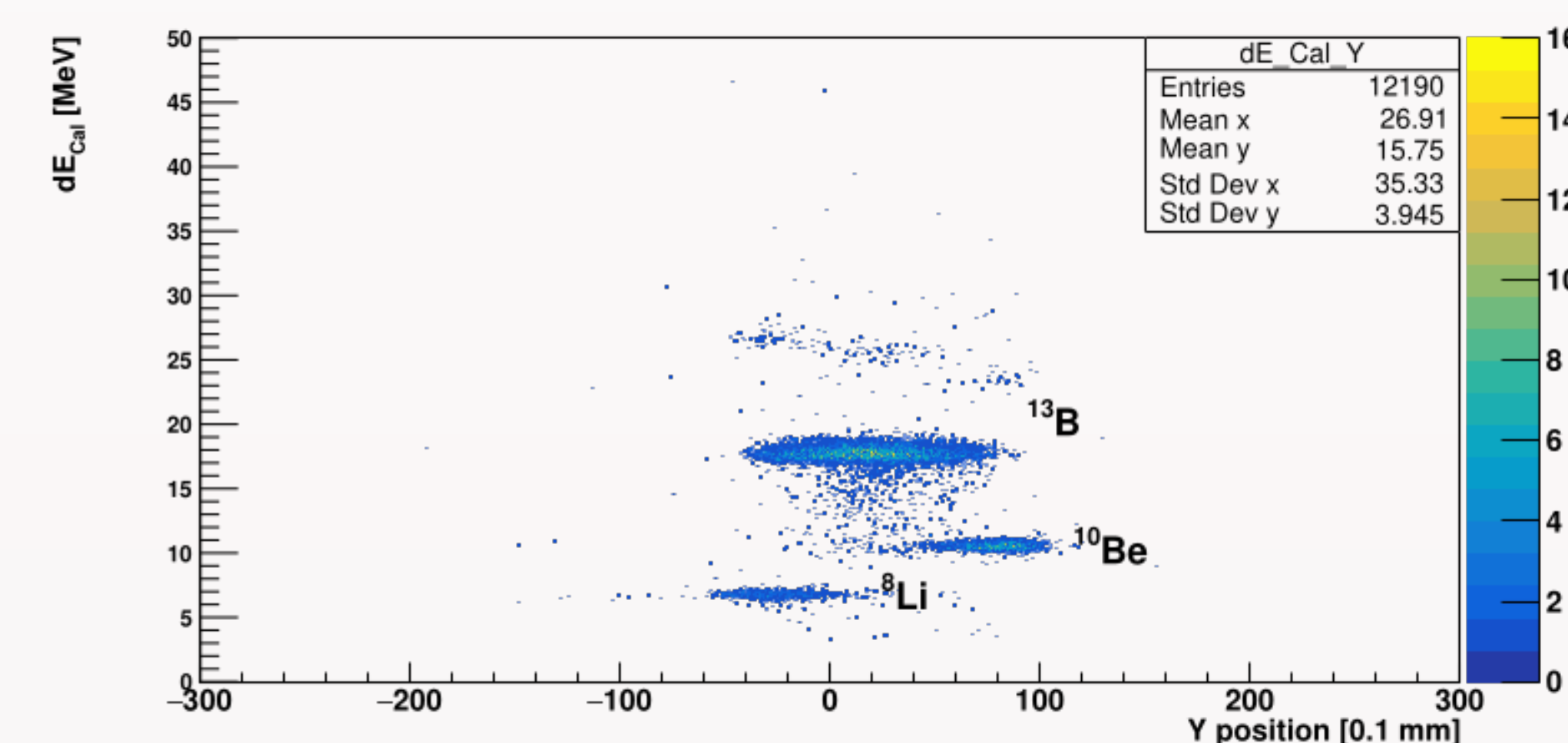
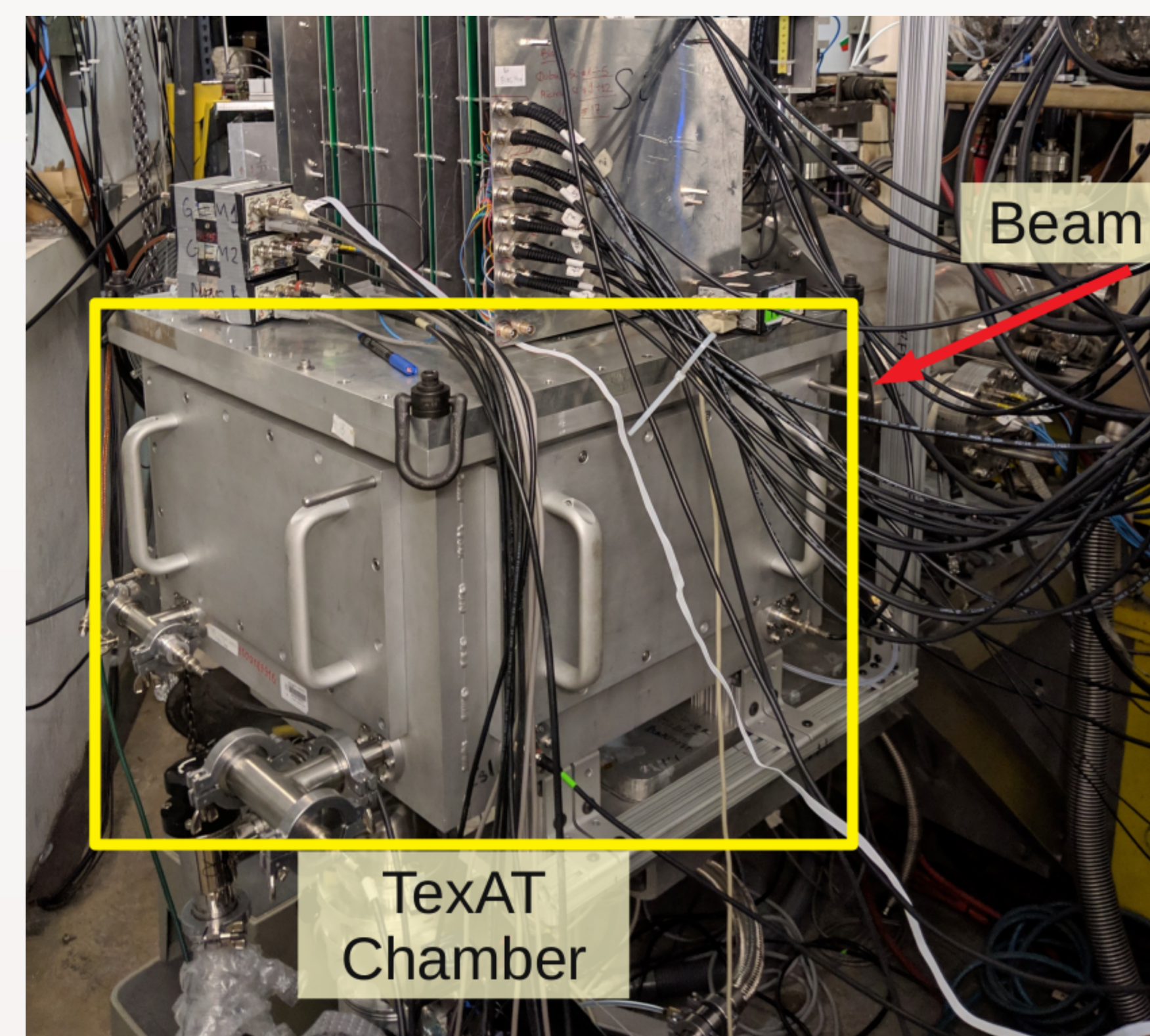
Data acquisition for TexAT is built around GET Electronics with GANIL DAQ.



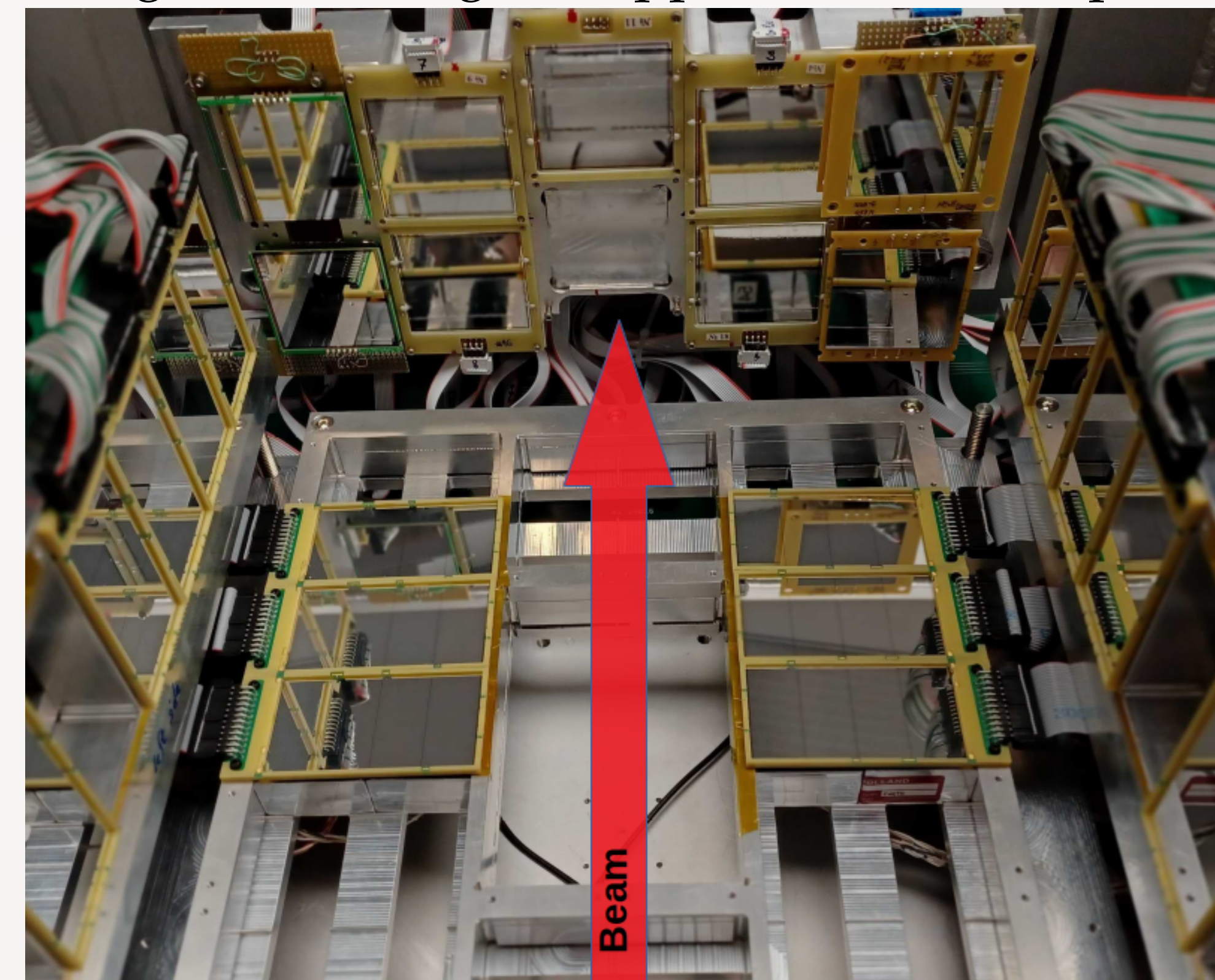
^{13}B BEAM TIME



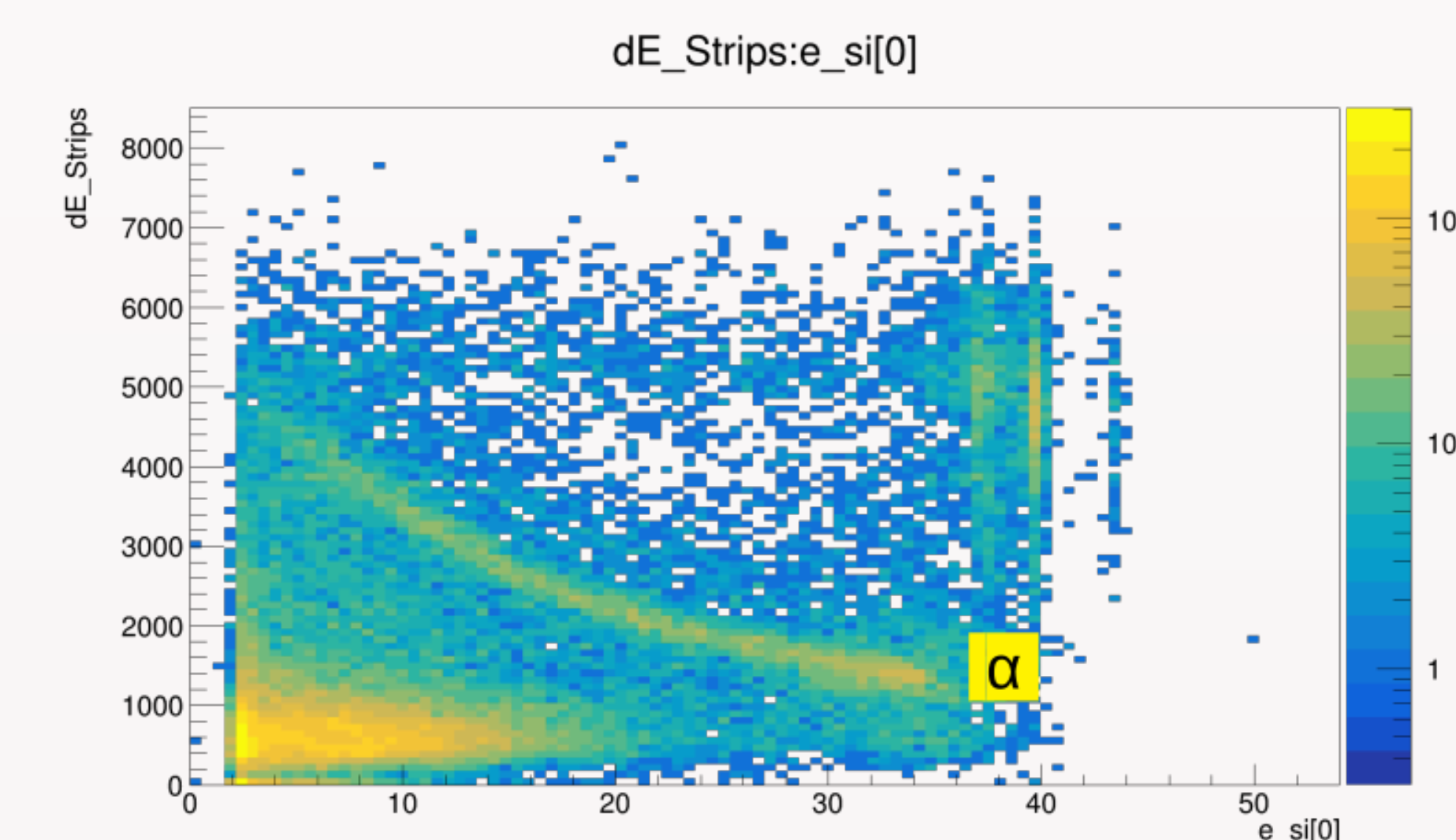
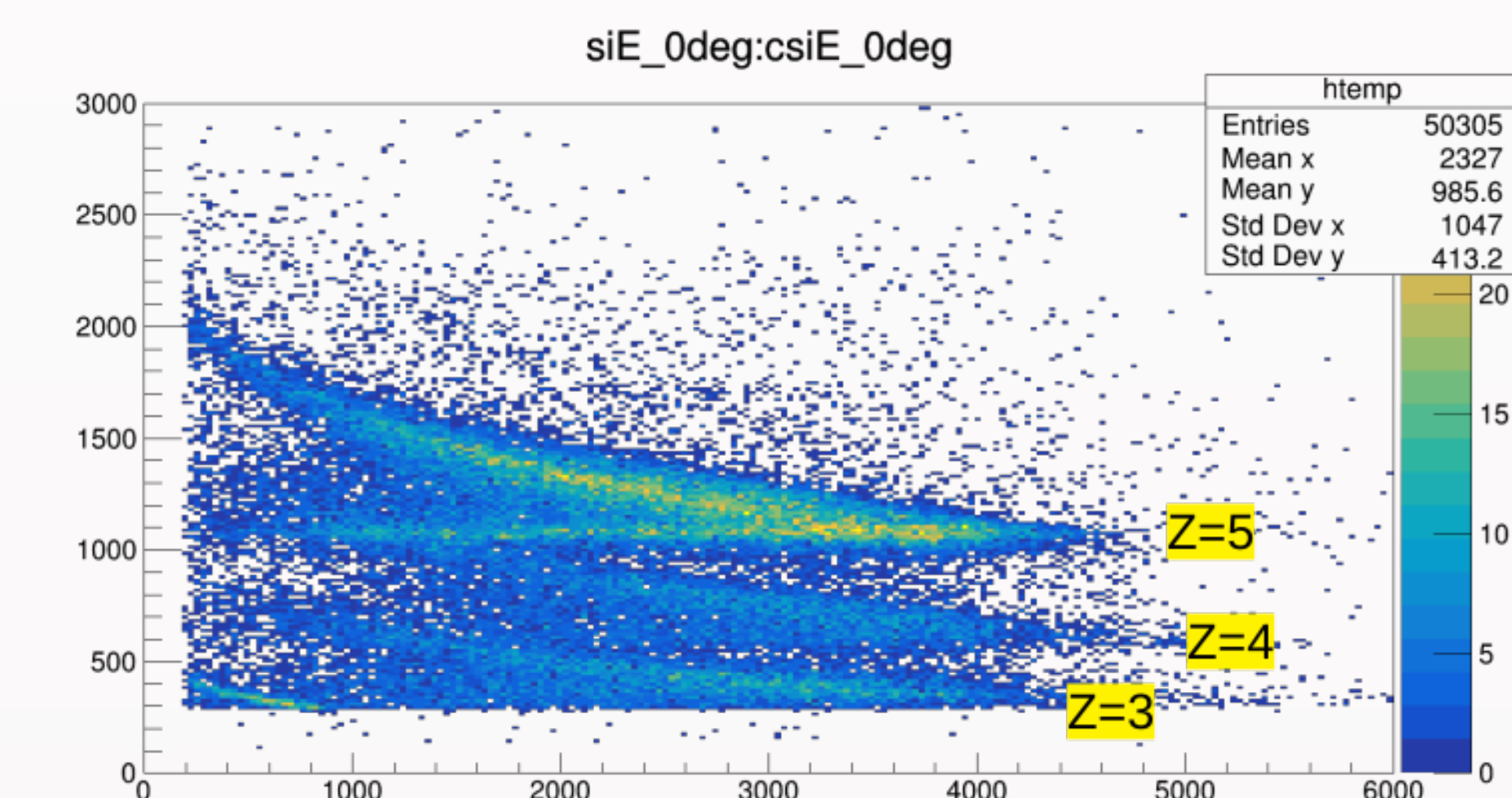
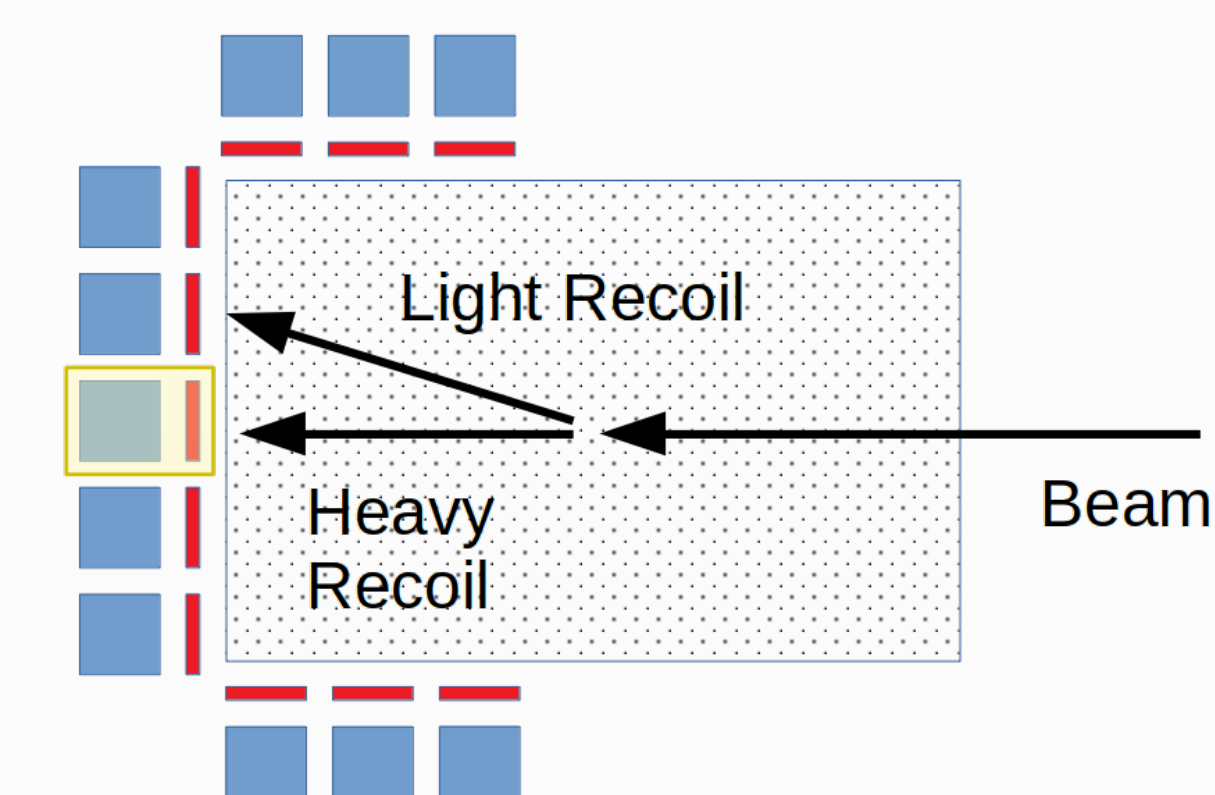
Primary: $^{15}\text{N}^{5+}$ 30 MeV/u
Target: ^9Be 1 mm thick
Product: ^{13}B 268 MeV



Using MARS we get 10^4 pps of ^{13}B @ 80% purity.



PROJECT PROGRESS



^3He and Be have been identified!

Next Steps:

- Fit tracks $\rightarrow \theta$ & vertex reconstruction.
- Reconstruct $^{12}\text{Be}^*$ energy spectrum.
- Compare $\frac{d\sigma}{d\Omega}$ to DWBA $\rightarrow C^2S$

ACKNOWLEDGMENTS

We would like to thank Brian Roeder, Antti Saastamoinen and the operations staff of the TAMU Cyclotron Institute for their help and making this research possible. This work is supported in part by the U.S. Department of Energy through award no DE-SC0018980.

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- [1]: T. Otsuka *et al.*, Rev Mod Phys **92**, (2020).
[2]: H.T. Fortune, PLB,**755** (2016) 351.

- [3]: E. Koshchiy, *et al*, arXiv:1906.07845.
See also: W. Liu *et al.*, Phys. Rev. C **105**, 034613